1.8 Redox Reactions

When an apple is cut, it turns brown after sometime. Do you know the reason behind this colour change? It is because of a chemical reaction called oxidation. We come across oxidation reactions in our daily life. For example 1) burning of LPG gas 2) rusting of iron



Fig. 1.4 Oxidation reactions in daily life

3) Oxidation of carbohydrates, lipids, etc. into CO_2 and H_2O to produce energy in the living organisms.

All oxidation reactions are accompanied by reduction reactions and vice versa. Such reactions are called redox reactions. As per the classical concept, addition of oxygen (or) removal of hydrogen is called oxidation and the reverse is called reduction.

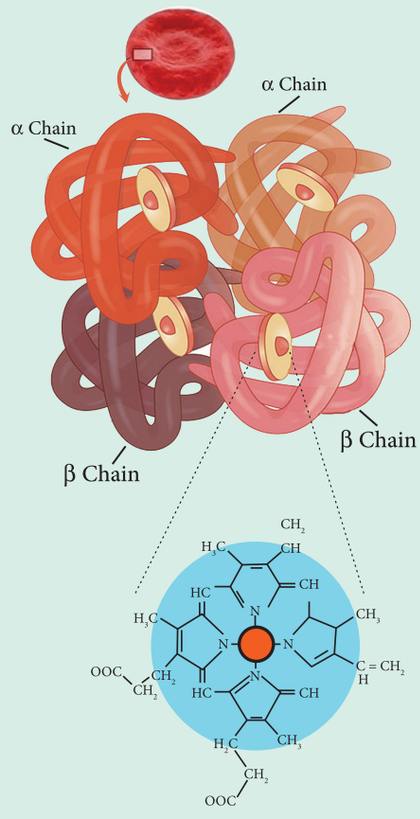
Haemoglobin and oxygen transport



Even a small amount of oxygen present in air leads to the rusting of iron, i.e. iron is oxidised to Fe^{3+}. But the Fe^{2+} present in haemoglobin which binds oxygen during transport of oxygen from lungs to tissues never gets oxidised. Do you know why?

The answer lies in the structural features of haemoglobin. Haemoglobin contains four sub units each with a porphyrin ring (heme) attached to the protein (globin) molecule. In this structure, the iron (Fe^{2+}) forms a co-ordination complex with an octahedral geometry. The four positions of the octahedron are occupied by porphyrin rings, fifth position is filled by imidazole ring of a histidine residue and the sixth position is utilized for binding the oxygen molecule. Generally the Fe^{2+} in heme is susceptible to oxidation. Since the Fe^{2+} ion in haemoglobin is surrounded by the globin protein chain that provides a hydrophobic environment, the oxidation of Fe^{2+} becomes difficult. However, 3% of haemoglobin is oxidised to methemoglobin (haemoglobin where the iron is present in Fe^{3+} state and oxygen does not bind to this) daily. The enzyme methemoglobin reductase reduces it back to haemoglobin.

Cyanide poisoning: While oxygen binds reversibly to haemoglobin, cyanide binds irreversibly to haemoglobin and blocks oxygen binding. As a result the transport of oxygen from the lungs to tissues is stopped. It leads to the quick death of the person.



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